

Magnetic Effects of Electric Current — MCQ Worksheet

Science • Chapter 12 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

- Q1.** The relationship between electricity and magnetism was first demonstrated by:
(A) Faraday (B) Oersted
(C) Ampere (D) Maxwell
- Q2.** Magnetic field lines around a bar magnet emerge from the:
(A) S pole (B) N pole
(C) Both poles (D) Neither pole
- Q3.** Magnetic field lines never:
(A) Form loops (B) Cross each other
(C) Touch poles (D) End
- Q4.** The closeness of magnetic field lines indicates:
(A) Weak field (B) Strong field
(C) No field (D) Reversed field
- Q5.** The magnetic field around a straight current-carrying conductor consists of:
(A) Straight lines (B) Concentric circles
(C) Parabolas (D) Random shapes
- Q6.** The right-hand thumb rule gives the direction of:
(A) Force (B) Magnetic field around a wire
(C) Electric field (D) Current only
- Q7.** Inside a current-carrying solenoid the magnetic field is:
(A) Zero (B) Uniform and strong
(C) Weak (D) Random
- Q8.** A solenoid behaves like a:
(A) Bar magnet (B) Single charge
(C) Capacitor (D) Resistor
- Q9.** An electromagnet is made stronger by inserting a:
(A) Wooden core (B) Plastic rod
(C) Soft iron core (D) Glass rod
- Q10.** Fleming's left-hand rule gives the direction of:
(A) Induced current (B) Magnetic field
(C) Force on a current-carrying conductor in a magnetic field (D) Magnetic flux
- Q11.** In Fleming's left-hand rule, the forefinger represents the:
(A) Force (B) Current
(C) Magnetic field (D) Velocity
- Q12.** An electric motor converts:
(A) Mechanical to electrical energy (B) Electrical to chemical energy
(C) Electrical to mechanical energy (D) Light to electrical energy
- Q13.** In a DC motor, the device used to reverse current direction in the coil every half rotation is the:
(A) Brushes (B) Split-ring commutator
(C) Battery (D) Switch
- Q14.** Electromagnetic induction was discovered by:
(A) Oersted (B) Maxwell
(C) Faraday (D) Edison

Q15. An electric generator works on the principle of:

- (A) Heating effect
- (B) Magnetic effect of current
- (C) Electromagnetic induction
- (D) Chemical effect

Q16. AC current in India has a frequency of:

- (A) 10 Hz
- (B) 50 Hz
- (C) 60 Hz
- (D) 100 Hz

Q17. Fleming's right-hand rule gives the direction of:

- (A) Force on a conductor
- (B) Induced current
- (C) Magnetic field around a wire
- (D) Velocity

Q18. In a domestic 3-wire system, the neutral wire is normally at:

- (A) 220 V
- (B) 0 V
- (C) 110 V
- (D) Negative voltage

Q19. The colour code for the earth wire is generally:

- (A) Red
- (B) Black
- (C) Green
- (D) Yellow

Q20. Short-circuit occurs when:

- (A) Live and neutral wires touch directly
- (B) Earth wire is loose
- (C) Voltage falls
- (D) Fuse blows

Q21. The principle behind the working of an electromagnet is the:

- (A) Heating effect of current
- (B) Magnetic effect of current
- (C) Chemical effect
- (D) Light effect

Q22. A current-carrying conductor experiences maximum force in a magnetic field when current is:

- (A) Parallel to field
- (B) Anti-parallel to field
- (C) Perpendicular to field
- (D) At 30° to field

Q23. Galvanometer detects:

- (A) Voltage only
- (B) Resistance only
- (C) Presence of small currents
- (D) Magnetic field strength only

Q24. A coil rotated in a magnetic field generates AC because:

- (A) Field reverses
- (B) Direction of induced EMF reverses every half rotation
- (C) Coil resists
- (D) Magnets repel

Q25. An MCB (Miniature Circuit Breaker) acts like a:

- (A) Voltage stabiliser
- (B) Fuse with reusable mechanism
- (C) Battery
- (D) Capacitor